

REED PUTNAM PRACTICE · FALL 2026

Exam-Day Strategy

Week 9 · triage, write for points, sleep

Thu Dec 3 / Fri Dec 4

The 2026 format

Saturday, December 5

Four 90-minute sessions with breaks (was two 3-hour sessions). Twelve problems, A1–A6 and B1–B6, 10 points each, across the four sessions [SPLIT TBD: confirm the problem count and session split from the proctor's instructions].

- > Easiest first within each half: A1, A2, B1, B2 are written to be solved.
- > No leaving and re-entering. Bring blue/black pens and your Putnam ID number.
- > No calculators, notes, books, drawing tools; phones and watches silenced, put away.
- > ROOM TBD, 20 minutes early. Sleep by 9 pm; breakfast; water and a snack.

Triage

The first ten minutes of every session

Read **every** problem before writing; one line each: object, what is asked, which week it smells like. Start with the one you **understand best**, not the smallest number.

- A1/B1 take 30–45 min. Nothing after 20? **Re-read word by word**, try $n = 1, 2, 3$, extremal case.
- Per session: 10 triage · 35 best · 25 next · **15–20 write-up**. Unwritten = 0 points.
- No idea for 15 min, easier waiting? Abandon (mark the time); return once the easier ones are written.

Writing for points

Scores are 0, 1, 2 or 8, 9, 10

Complete (with cosmetic slips) or not; partial progress is a point or two. So: **fewer** problems, fully written.

- > Answer first, in closed form, then prove it. “Determine all” needs **both directions**.
- > Full sentences, every quantifier visible; a page of equations is not a proof.
- > The key step is the one you must **not** hand-wave: write the line that shows it.
- > Name the tool as you use it; cross out dead ends with one line, never leave two arguments.
- > **Lose 8**: one direction only; a fact with unchecked hypotheses; key step not on paper.

Technique checklist

- > Wk 1 pigeonhole, extremal, induction. “Among any N , some two”: choose the boxes or take the extreme object; “for all n ”: induct, strengthen.
- > Wk 2 invariants & parity. Moves, “can one reach”: a preserved or monotone quantity (parity, sum mod m , coloring).
- > Wk 3 polynomials. Integer coefficients, roots: $(a - b) \mid P(a) - P(b)$, Vieta, degrees and leading terms, factor over $\mathbb{C}$.
- > Wk 4 number theory. “Divisible by,” primes, n th powers: mod a small prime, orders, v_p (min valuation occurs twice), descent.
- > Wk 5 combinatorics. “How many,” “evaluate”: do $k = 0, 1, 2, 3$, then bijection, double counting, recursion, generating function.
- > Wk 6 inequalities & analysis. “Show $\geq$,” “largest M ”: AM–GM, Cauchy–Schwarz, convexity; equality case = the example; squeeze.
- > Wk 7 linear algebra. Matrices, rank, orthogonal vectors: integer det, $\det(AA^T) = (\det A)^2$, eigenvalues and trace, rank–nullity.
- > Wk 8 probability & games. “Random,” “expected,” “two players”: linearity, symmetry, pairing, stealing, small positions.

Now you

35 minutes.

Work alone. Full write-ups. Go.